Learning Distributed Algorithms from Algorithmic Teachers

Research collaboration (author order pending)

23 October 2026

Unpublished integrated research draft, 23 October 2026. Author list and order remain to be agreed. Historical contributions are attributed by Git provenance.

Abstract

Learning a distributed algorithm can mean predicting its actions, reproducing its transitions, solving its task, matching its running time, or generating the same probability law over solutions. These objectives need not coincide. We investigate distributed graph coloring through seven explicitly scoped algorithmic teachers and shared neural policies. A historical five-seed architecture study finds that normalized node/color networks, DeepSets palette scorers and local palette Transformers each complete all 1,520 held-out evaluations, while aggregation and representation controls expose failures. A complementary seven-teacher study shows that almost-perfect local action accuracy can coexist with substantial autonomous failure. We integrate the three palette architectures with every native teacher, add small and no-communication controls, and train 70 balanced short-budget replicas. We separately audit teacher safety and termination, compare independent scalar specifications on 103 small graph representatives, and distinguish supplied arithmetic and schedules from learned decisions. Statistical reconciliation aligns the compared output laws and shows agreement between two procedures on six identical archival banks. Fresh known-law controls reveal a large power difference for localized alternatives. Our evidence supports a scoped separation between functional generalization and fidelity to a teacher's complete solution law. Missing historical collaborator weights prevent a complete numerical replay of that cohort; no conclusion imputes those artifacts or turns a nonrejection into equality.

1 Introduction

A distributed coloring algorithm is an executable local rule embedded in a communication model. A neural learner trained from its trajectories must represent that rule, survive the states induced by its own decisions, and remain meaningful when the number of neighbors or colors changes. A proper coloring at the end of an execution is one useful outcome, but it leaves several questions unanswered. Did the network make the same decisions? Did it use the same information? Did it preserve the teacher's progress guarantee? Did it sample the same coloring distribution?

These questions become sharper when teachers differ. Random trials use private randomness to resolve conflicts; priority rules exploit distinct identifiers; color-wise preferences compare several possible colors; auxiliary color reduction uses finite-field representations; speculative programs repair temporary conflicts; and sequential greedy coloring assumes a supplied order. A common

neural interface can compare their local decisions only if those differences remain explicit. Equal parameter counts do not compensate for different supplied information or computational schedules.

We study five notions of learning: **action fidelity**, agreement on common teacher states and private inputs; **transition fidelity**, agreement of updated states; **functional validity**, producing a proper complete coloring; **complexity fidelity**, preserving an appropriate teacher-specific cost; and **distributional fidelity**, agreement of the probability laws of complete solutions on a fixed labeled graph. The central question is how these notions relate across teacher, architecture and evaluation regime.

The project combines two complementary lines. The degree-generalization studies emphasize variable palettes, neighborhood aggregation, conditional safety/progress and full labeled output distributions. The native-teacher studies emphasize distinct algorithms, supplied phase interfaces, local imitation and autonomous size transfer. Their results are not pooled as if they used one protocol. Instead we reconstruct the definitions, preserve their separate cohorts and add a balanced integration experiment.

Our strongest conclusion is deliberately qualified: some learned local policies generalize functionally while differing from their teacher in observable full-output laws. This is not a claim that every successful model must differ, or that every difference can be detected at our sample sizes. Symmetry can force equality on particular graphs, and a valid coupling can certify approximate agreement on a fixed graph. A coherent account must accommodate all of these outcomes.

1.1 Contributions and their status

First, we specify a teacher/architecture interface preserving seven native action alphabets and their supplied information. E1, DeepSets and local palette attention are represented for each teacher, with a controlled short-budget experiment and two additional baselines. This is a framework and empirical study, not seven newly invented distributed algorithms.

Second, we provide conditional proofs for the coded exact teacher rules, an independent scalar transition audit and explicit limits of finite testing. A repaired input-domain guard has a saved failing regression. We separate correctness of a mathematical teacher from correctness of an arbitrary trained numerical policy.

Third, we reconcile the statistical objects before comparing findings. Matched-bank replays preserve original statistics; fresh known-law controls quantify a finite-sample sensitivity difference. Existing factorial data add observable-event bounds and size/degree interactions without asserting a monotone law for total variation.

Fourth, the repository supplies an evidence ledger, complete scientific audit, frozen checkpoint manifests, historical negative results, executable reproduction commands and remotely verified recovery artifacts. Historical collaborator checkpoint-level reproduction remains explicitly incomplete.

2 Related work

Neural algorithmic reasoning studies learned execution of algorithmic computation [Veličković and Blundell, 2021]. CLRS supplies a common benchmark spanning classical algorithm families [Veličković et al., 2022]. Our question focuses on repeated local decisions in specified distributed interpreters, including randomness and stopping semantics, and evaluates a solution law as well as task success. We do not claim that trajectory imitation or algorithmic generalization is new.

Distributed locality restricts the information available within a bounded communication radius [Linial, 1992]. Connections between local algorithms and GNN expressiveness clarify the importance of inputs, depth and aggregation [Sato et al., 2019, Loukas, 2020]. Those results do not imply that a particular optimized network generalizes correctly. Our controlled observations and conditional proofs concern explicit architectures and teachers, rather than a universal GNN expressivity theorem.

Deep Sets gives a principled shared-function/pooling construction for sets [Zaheer et al., 2017], while attention supplies content-dependent interaction [Vaswani et al., 2017]. Here both act over each node's palette; graph communication remains a neighborhood operation. Our positional palette features retain color-name information, so these models must not be described as fully color-permutation invariant.

Imitation learning encounters a state-distribution shift when learner actions affect later observations [Ross et al., 2011]. We use teacher trajectories and counterfactual COMMIT labels, and evaluate autonomous rollouts. This is not equivalent to on-policy dataset aggregation. Work on asynchronous GNN inference also warns that synchronized behavior need not survive changed execution semantics [Solodova et al., 2025]; this paper does not test asynchronous correctness.

The teachers draw on randomized coloring [Luby, 1993, Johansson, 1999], priority coloring [Jones and Plassmann, 1993], polynomial and iterative reduction [Linial, 1992, Kuhn and Wattenhofer, 2006, Barenboim and Elkin, 2008], color-wise preference trials [Schneider and Wattenhofer, 2010] and speculative repair [Gebremedhin and Manne, 2000]. We state the implemented specialization rather than transferring the full complexity guarantee of each cited paper.

Kernel two-sample tests compare probability laws through MMD [Gretton et al., 2012]. Characteristicness is an identifiability property, not a finite-sample power guarantee. Random-permutation calibration and familywise correction require their own assumptions [Hemerik and Goeman, 2018, Holm, 1979]. Our honest witness omnibus is a specified empirical instrument; no general minimax optimality or universal dominance is claimed. This is a scoped primary-source review, checked on 23 October 2026, rather than an exhaustive novelty determination.

3 Formal model and notions of fidelity

Let $G = (V, E)$ be a fixed finite simple undirected labeled graph, $n = |V|$, and let C_v be the allowed colors of vertex v. A state contains a partial coloring $q_v \in C_v \cup \{-1\}$ and any explicitly declared auxiliary variables. A proper partial state has no equal fixed colors across an edge. For permanent-color trial teachers,

$$A_v(q) = C_v \setminus \{q_u : u \in N(v), q_u \geq 0\}.$$

In the local-palette studies, $C_v = \{0, \dots, d(v)\}$. Native degree-three comparisons use a common $C_v = \{0, 1, 2, 3\}$ even at vertices of lower degree. The two settings coincide on cubic graphs and generally differ on irregular ones. No common maximum-degree input is supplied to the local-palette E1 policy. Some native adapters explicitly use the known bound 3.

A protocol step maps a current state and private input to participation B, candidate C, confirmation J, an auxiliary choice, or a repair mark, according to its native phase. A permanent fixation is applied only at a pending, permitted participating node with a candidate and positive COMMIT.

Palette range, pending-node gating, native phases and schedules are supplied mechanisms; availability and conflict decisions are learned unless explicitly identified as an exact-teacher baseline.

For a frozen teacher T and policy M, action fidelity evaluates their decisions on identical input states and explicit random arrays. Transition fidelity also evaluates the resulting states. Functional success is the event that all vertices receive an allowed color and every edge has different endpoint colors. These three predicates need not coincide. A policy may produce another valid solution after its first action disagreement; an inaccurate decision may instead propagate to a later conflict.

For cutoff R, write S_R for proper completion and $P^G_{T,R} = \mathcal{L}(X_G \mid S_R)$, where $X_G = (c_1, \dots, c_n)$. Vertex and color names remain fixed. A joint terminal law instead observes colors, partial outcomes and status at R. A quotient law canonicalizes color names. These are different estimands. If $P(S_R) = 0$, the conditional solution law is undefined and cannot be estimated by resampling until success.

Total variation is $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. A statistically significant fixed-event difference therefore gives a TV lower bound. For any valid coupling (X, Y), $\mathrm{TV}(P, Q) \le P(X \neq Y)$ instead gives an upper bound. Large disagreement under a poor coupling supplies no lower bound. In particular, shared-randomness action fidelity and solution-law fidelity are distinct questions.

4 Algorithmic teachers

Table 1. Implemented teachers and computational scope. All native comparisons below have degree$\le$ 3 and global K4. Their exact references and proof assumptions are expanded in AppendixA and the teacher-audit document.

Teacher	Local decision	Supplied information and cost convention
LUBY	Bernoulli 1/2 participation; random available rank; reject equal neighboring proposals	Two exchanges per trial; no IDs needed
JOHANSSON	Always-active random available rank; reject equal neighboring proposals	Common K4; two exchanges per trial
JP	Pending strict ID maxima take first free color	Fixed distinct IDs; synchronous native rounds, capped 128
COLOR_TRIALS	Strict discrete preference winner per color; keep lowest winning color	Preferences 0–3, availability exchange; repeated subroutine
BE_BASE_D3	Linial field-point reductions then KW recoloring	Phase clock, polynomial evaluation and receiver-relative equalities supplied
SPECULATIVE_GM	Concurrent block assignment, same-slot detection, serial repair	Four balanced blocks and writable schedule supplied; intermediate conflicts allowed
GREEDY	One permitted vertex takes first free color	Sequential ID-order schedule supplied

The word LUBY denotes the explicit coin-participation random-trial rule. It does not mean that this repository implements every derandomization or PRAM component of Luby's work. JOHANSSON uses a larger common initial palette at lower-degree vertices; the original analysis permits larger initial lists, but the output law differs from the smallest local-list initialization. Repeated COLOR_TRIALS substitutes a fixed global degree bound for its neighborhood bound and omits RankingTrials. BE_BASE_D3 is the degree-three base construction with two explicit Linial steps, not the high-degree recursive algorithm. GM and greedy retain schedules that would require a separate distributed implementation if no scheduler were supplied.

The four parallel trial/priority interpreters stop at 128 native steps. BE uses its auxiliary phases; GM and greedy run their supplied schedules. The historical canonical E1 behavior experiment uses cutoff 256 and continues after a collision. We retain these differences rather than calling all reported native steps a common LOCAL running time.

5 Neural policies

5.1 E1 NORMALIZED

For each vertex/color pair define pending status u_v and ownership $o_{vc} = \mathbf{1}[q_v = c]$. Shared learned functions send a nonnegative message and form a local state,

$$m^1_{vc} = \phi_1(u_v, o_{vc}), \qquad h_{vc} = \psi_1(u_v, \max_{w \in N(v)} m^1_{wc}).$$

Availability is learned as $\alpha_{vc} = \sigma(a(h_{vc}))$. The ordered prefix $S_{vc} = \sum_{j<c} \alpha_{vj}$ and $D_v = \max(\sum_j \alpha_{vj}, .5)$ give normalized endpoints $L_{vc} = S_{vc}/D_v$, $U_{vc} = (S_{vc} + \alpha_{vc})/D_v$. A shared scorer reads h, α, L, U, r^c_v; the candidate is the maximum scoring palette slot. Occupied colors are not removed by an exact mask.

The second exchange represents pending/ownership/participation/offer bits. A learned COMMIT head reads the candidate slot after neighbor MAX. Its prediction is applied without an exact safety veto. Canonical E1 has 16,035 parameters with hidden/message dimensions 64/32. This count is independent of palette size, but messages, memory and computation grow with the palette; each directed edge carries 32K floats per message stage in the direct implementation.

5.2 DeepSets and local palette attention

Both variants retain learned neighborhood messages, availability and COMMIT. DeepSets replaces ordered-prefix scoring with shared per-slot features and a mean of learned palette features. LocalPaletteTransformer projects slots to dimension 16 and uses two-head self-attention with a 32-dimensional feedforward layer and zero dropout. Its attention is within one vertex's palette, not between arbitrary graph vertices.

Their slot inputs include learned availability, normalized color position, inverse palette size and private randomness. Prefix-free is therefore not position-free. All three models are vertex-equivariant in the ideal shared-rule, invariant-aggregation model, while ordered prefixes and positions generally break color-name permutation symmetry. Floating-point backend probes support finite implementations but do not prove universal numerical equivariance.

5.3 Native teacher adapters and baselines

The common native interface preserves Pedro's two-hop rooted representation and exact action alphabets. LUBY/Johansson/greedy use the unextended core. JP, ColorTrials, BE and GM retain supplied native context residuals. The E1 version adds 3,361 parameters in those cases; analogous context is retained for the other palette architectures. ColorTrials selects its final candidate after the second message. BE-Linial predicts a valid field point and computes the polynomial value through declared local arithmetic. The DeepSets and Transformer masks use the actual field/palette size, including mixed field 7/13 examples.

Two new controls use the same supervision and native action interfaces. The **small** model reduces hidden/message dimensions to 8/4 inside the normalized family. The **no-communication** control erases received neighbor states, relations and masks while retaining own state and supplied native permissions. It is an information ablation with the same large core, not a claim of parameter minimality. Exact teachers are zero-trainable-parameter task baselines. The earlier direct-action 36,719-parameter GNN remains a historical comparator; it uses different representation/loss choices.

No architecture is called minimal merely because it has fewer parameters. We distinguish a mathematical information limitation, sufficiency within a tested architecture family, and finite empirical success.

6 Training and experimental design

6.1 Three distinct training cohorts

The historical mechanism study trains five seeds 41000–41004 per primary architecture on 800 cubic N40 graphs, three teacher trajectories,120 epochs, Adam at .001 and batch 128 round examples. The full-graph E1 loss combines participation BCE, candidate cross-entropy, and equal half-weights for own and counterfactual COMMIT supervision. During ten warmup epochs COMMIT uses teacher proposals. Afterwards exact labels are recomputed for the proposals actually queried; hard proposals separate gradients into decision modules. Training states remain teacher-generated. Rollout evaluation supplies the autonomous-state test.

Pedro's native comparisons instead use five seeds 0–4, common immutable pools of 800 cubic N40 graphs with three trajectories,17,760 updates and batches of 1,680 eligible root decisions. The direct GNN uses native-action losses. Its E1 follow-up uses 1,480 warmup updates and native own/counterfactual confirmation labels. Star-local and full-graph intervention samplers are different. The native follow-up also changes BE's auxiliary representation. Consequently its architecture comparison measures a training/representation package, not an isolated architectural intervention.

The integrated extension uses 24 new cubic N40 graphs, one trajectory each, eight validation graphs and two new seeds 73000–73001. Every teacher has the same five architecture cells. Timing pilots selected 500 common updates of 128 root decisions, with 50 warmup updates, before scientific pools were generated. This exposes 64,000 sampled root decisions per model. It is intentionally a short-budget study, not a repeat of 29,836,800-decision native training. All 70 final checkpoints are retained without validation promotion or a second training budget after observing outcomes. Compatible shared message/COMMIT initial weights are preserved across the three large palette models within a seed; the small family has different dimensions.

The two initialization replicas share teacher-specific data pools. They do not replicate data gener-

ation, and repeated rollout tapes do not create additional independently trained models. Hyperparameters, parameter counts, source hashes, training time, data manifests and checkpoint digests are recorded in `results/integration/multiteacher/`.

6.2 Evaluation and units of replication

The full-budget architecture cohort contains 304 common graph/tape cases per model, giving 1,520 cases for five seeds of one architecture. It covers degree and size transfer, irregular/heterogeneous families, cliques and structured controls. The native cohort contains 396 cases per model through N10,240. Its E1 follow-up uses already-published cases and is exploratory. Five seed outcomes on one graph are not five independent graph instances.

The new native extension uses five independent graph specifications: regular N16, irregular N40, regular N160, path 16 and K4, excluding training/validation labeled fingerprints. Every exact teacher and checkpoint produces 128 new independent executions on each graph. Full color vectors, statuses, native steps, failure labels and random seeds are archived. The graph ensemble is small and fixed; no confidence interval treats it as a representative sample of all graphs.

Participation, candidate and COMMIT accuracy use common teacher states and explicit private inputs. Native COMMIT labels retain their own semantics: ColorTrials preference rejection is not identical to a physical collision; GM assignment can deliberately create simultaneous conflicts before repair. Per-phase action accuracy, component denominators and GM mark counts are retained so WAIT or negative-mark imbalance cannot alone establish learning.

Functional evaluation applies the model's own decisions, with native permissions but no oracle conflict veto. We report invalid outputs and incomplete outcomes alongside successes. Mean native steps over all attempts can be reduced by early failure; efficiency comparisons therefore also use successful outputs and remain within one teacher. Parameter count, training CPU time and native steps measure different costs. No deployment latency or cross-teacher LOCAL speed ranking is inferred.

6.3 Distributional inference

The primary observation is an independent complete run. For the historical local-list studies, V2 splits the valid completed prefix into selection and evaluation subsets, fits bounded witnesses on selection only, and calibrates their joint maximum with a full-vector MMD component on evaluation rows. Every permutation acts jointly on complete row labels; vertex columns are never independently shuffled. Pooled scaling ignores evaluation source labels. The permutation count 19,999 gives minimum p .00005.

The factorial primary family has 720 graph/checkpoint hypotheses and the robust study 795. Event, irregular-graph, historical-replication and explanation families remain separate. For the new extension, all 350 planned teacher/model/ graph comparisons form one Holm family. Every model is compared with its own teacher's exact reference. A cell with fewer than four valid completed rows is untestable and assigned p 1 only for planned-family bookkeeping. It is not an equality result. Conditioning, failure frequencies and the native cause of termination remain visible.

Pedro's separate behavior experiment uses a mixture of Hamming kernels and terminal-status equality, plus family/size-stratified permutation aggregation. It also tests a color-name quotient and obtains coupling-based TV upper bounds. We audit its exact statistic and compare procedures on matched rows, without pooling its hypothesis family with our graph-specific families.

7 Conditional correctness and representation results

Proposition 1 (safety and progress are separate). For a proper partial state, candidates supported on available colors and exact rejection of simultaneous equal neighboring offers preserve properness. This alone does not imply a uniform progress or termination bound.

Proof. A new color avoids old fixed neighbors by availability, and two adjacent equal proposals cannot both be confirmed. This proves safety by induction. In an all-free K-clique, suppose every node proposes color 0 and participates independently with probability 1/2. A node confirms only when it is the unique participant, so the probability that any node advances is $K/2^K$. Waiting for the first confirmation has mean $2^K/K$. A policy that always waits never terminates. Safety thus supplies no constant progress bound. The example concerns specified policies, not every learned checkpoint. □

Proposition 2 (exact teacher guarantees on the declared domain). Under fresh independent ideal private randomness and the hypotheses in Table 1, the uncapped LUBY, JOHANSSON and repeated ColorTrials teachers preserve properness and terminate almost surely, with O(log n) expected native rounds at degree$\leq$ 3. JP is safe and terminates within n uncapped native rounds. BE_BASE_D3 deterministically reaches four auxiliary colors in 15 native steps. Greedy and synchronized speculative GM terminate under their supplied schedules and produce proper final colorings.

Proof sketch; complete arguments in AppendixA. Availability satisfies $|A_v| \geq d_U(v) + 1$. LUBY has conditional progress at least 1/4 using the independent candidate product and Jensen's inequality. For JOHANSSON each pending neighbor has at least two available colors, giving the sufficient degree-three bound 1/8. For ColorTrials, the event that a vertex has preference 3 for a free color and all at most three competitors have preference$\leq$ 2 has probability at least 27/256. A uniform positive conditional progress boundρ gives $P(T > t) \leq n(1-\rho)^t$ by iterated conditioning and union bound. JP's eligible maxima form an independent set and include a maximum pending ID. Polynomial reductions preserve proper auxiliary colors; KW uses disjoint four-color blocks. Sequential first fit and the last-repaired endpoint argument complete the remaining cases. Finite caps are distinct from these uncapped guarantees. □

Proposition 3 (a finite-type sufficient COMMIT certificate). For the canonical MAX architecture, assume nonnegative deterministic shared messages, zero empty aggregation, legal candidate-slot gathering and pending/participation gating. A participating pending center's neighbor slot has six possible legal types. If the COMMIT logit has the correct strict sign on all 64 subsets of these types, it implements exact rejection at arbitrary degree and palette in real arithmetic.

Proof. Componentwise MAX is commutative, associative and idempotent, so only the set of present types matters, not multiplicities. The gather retains only the own candidate slot. Fixed ownership of that color and simultaneously participating equal offers are the two forbidden types. The 64-set sign table is therefore sufficient. It does not establish candidate diversity or termination. Native continuous-ID/context extensions are not automatically covered by this alphabet. Interval computations and floating-point error assumptions are documented separately. □

Proposition 4 (a scoped information limitation). A deterministic policy given only its own color/pending status and private draw cannot reproduce all arbitrary-state first-free decisions without neighbor information. Further, in the memoryless current-state interface, some random-trial COMMIT decisions require information beyond radius one.

Proof. A pending center with the same own inputs but a fixed neighbor of color 0 versus 1 has different first-free colors; its identical local input cannot prescribe both. For the radius example use path 0–1–2, with 0 and 1 pending and 2 fixed at 1 in one state and 2 in the other. Let both pending vertices participate, with candidate draws .3 at 0 and .4 at 1 in common K4. Vertex 0 proposes 1 in both states. Vertex 1 proposes 2 in the first state and 1 in the second, changing vertex 0's COMMIT. Its radius-one initial states/private inputs agree. This lower bound assumes no previously stored neighbor availability or supplied free-color flags. It is not a lower bound for every coloring algorithm, every persistent-memory protocol or every scheduling model. □

Proposition 5 (clique conditional-law equality). On K_n with exactly n colors, any vertex-equivariant protocol with exchangeable private input and positive probability of proper completion has the uniform conditional law on the n! proper labeled color vectors.

Proof. Equivariance and exchangeability give a vertex-invariant trajectory law by induction. The proper-completion event is invariant, so conditioning preserves this invariance. Vertex permutations act transitively on all proper n-color vectors. All conditional masses are therefore equal. Different candidate rules can satisfy the same conclusion. This is not proved by finite nonrejection and need not hold on other graphs. □

Proposition 6 (observable and coupling bounds). A simultaneous interval $[l, u]$ for $P_M(A) - P_T(A)$ gives $\mathrm{TV}(P_M, P_T) \geq \max(0, l, -u)$. A valid coupling gives the upper bound $\mathrm{TV}(P_M, P_T) \leq P(X_M \neq X_T)$.

Proof. The first inequality follows by restricting the supremum defining TV to the eventA and propagating its confidence statement. For the second, the difference of probabilities of any event is bounded by the chance that the coupled outcomes disagree. Taking the supremum proves the bound. Neither compares TV distances across different-dimensional graph spaces. □

8 Learning across teachers: historical evidence

Table 2. Pedro's published full-budget native comparison. Each entry is successful graph/tape/checkpoint evaluations out of 1,980. These aggregate counts are preserved historical evidence; their original weights and raw output banks are missing here. The follow-up reuses known test cases.

Teacher	Direct-action GNN	E1 / declared native adaptation
LUBY	1,857	1,980
JOHANSSON	1,919	1,980
JP	1,635	1,631
COLOR_TRIALS	1,980	1,980
BE_BASE_D3	872	1,980
SPECULATIVE_GM	1,975	1,951
GREEDY	1,842	1,980

These results establish no universal teacher ranking. E1/adaptations improve the reported LUBY, Johansson, BE and greedy outcomes, leave ColorTrials at complete observed success, and do not improve JP or GM. BE's large change also changes auxiliary representation and training. Both positive and negative before/after comparisons matter.

The JP E1 follow-up has mean local action accuracy .999735 but succeeds in 1,631/1,980 autonomous evaluations. This directly illustrates accumulation and state-distribution effects; it does not identify a unique causal mechanism for every failure. The reported GM losses similarly warn that exact local validation on sampled teacher states need not guarantee all rare conflict marks in large autonomous executions.

9 Functional generalization and aggregation

Table 3. Full-budget historical architecture controls. Five independent training seeds share 1,520 held-out graph/tape/checkpoint evaluations per architecture; the fixed evaluation instances are blocked rather than treated as independent new graphs per seed.

Architecture / intervention	Reported principal outcome
E1 NORMALIZED	1,520/1,520 proper completions
DeepSets palette	1,520/1,520 proper completions
LocalPaletteTransformer	1,520/1,520 proper completions
Raw endpoint control	Lost progress in 346 evaluations
SUM aggregation	Stalled in 1,196 evaluations
MEAN aggregation	Produced 839 improper final colorings

The normalized/raw-endpoint intervention preserves scorer tensor shapes, initialization, data and update counts within seed, making it more controlled than the earlier raw-prefix architecture comparison. SUM and MEAN alter neighbor reductions and multiplicity handling. Their failures are evidence about those implementations and training conditions, not a proof that SUM or MEAN cannot represent any correct coloring policy.

The successful prefix-free models show that E1's original ordered prefixes are not necessary for observed functional generalization in this experiment. They still receive useful palette position information and learned graph messages. No conclusion says that generic attention alone supplies a distributed coloring algorithm. Constant parameter count also does not imply constant communication or inference cost.

10 Balanced teacher/architecture extension

Table 4. New balanced finite-cohort functional outcomes. Each model architecture/teacher cell pools two fixed initialization replicas, five fixed graphs and 128 independent runs per graph/checkpoint, totaling 1,280 attempts. The entries are counts of proper completions, not population confidence bounds or full-budget architecture rankings.

Teacher	E1	DeepSets	Transformer	Small	No messages
Luby	1,280	1,280	1,280	1,280	30
Johansson	1,280	1,280	1,280	0	0
JP	473	395	448	1	0
ColorTrials	1,280	1,280	1,280	640	0
BE (base d 3)	232	268	255	1	8
Speculative GM	1,156	1,170	1,155	444	0

Teacher	E1	DeepSets	Transformer	Small	No messages
Greedy	1,280	1,280	1,280	1,280	0

Figure 1 displays the same finite-cohort percentages. Individual seed/graph records, failure types and native steps are available in `functional.csv` and the original hashed banks. The exact reference uses one 128-run bank per teacher/graph shared across its model comparisons; it is not counted twice to create extra reference data.

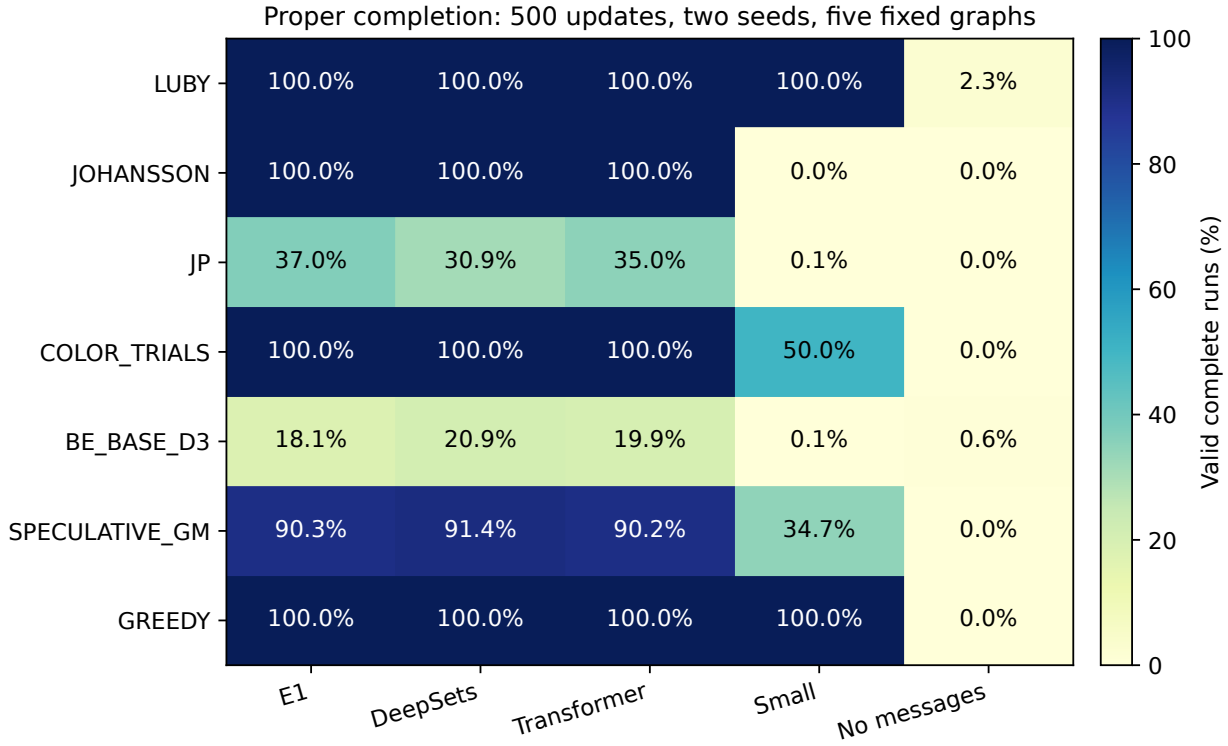


Figure 1: Proper-completion percentages on the balanced short-budget cohort. Every architecture receives 500 updates; each cell retains all failures and both training seeds.

All three large palette architectures completed 1,280/1,280 attempts for each of LUBY, JOHANSSON, COLOR_TRIALS, GREEDY. At the same short budget, JP and BE retained substantial failures, while speculative GM was intermediate. These differences compare specified training/representation packages, not intrinsic teacher complexity. Per-teacher architecture differences and both replicas remain in the table; there is no universal winner.

The short budget and small training graph sample limit interpretation. A failed cell can reflect optimization or coverage, and does not prove a representational impossibility. A successful small model establishes finite sufficiency within its family; it does not prove parameter minimality. The no-communication control retains native permissions, so its successes cannot be generalized to a scheduler-free anonymous setting.

Figure 2 compares common-state local validation accuracy and autonomous functional validity for all 70 separately trained models. Local validation contains different phase/class mixtures across

teachers, so between-teacher horizontal comparisons require the per-phase denominators. Each plotted point remains one checkpoint; rollout repeats are not training replicas.

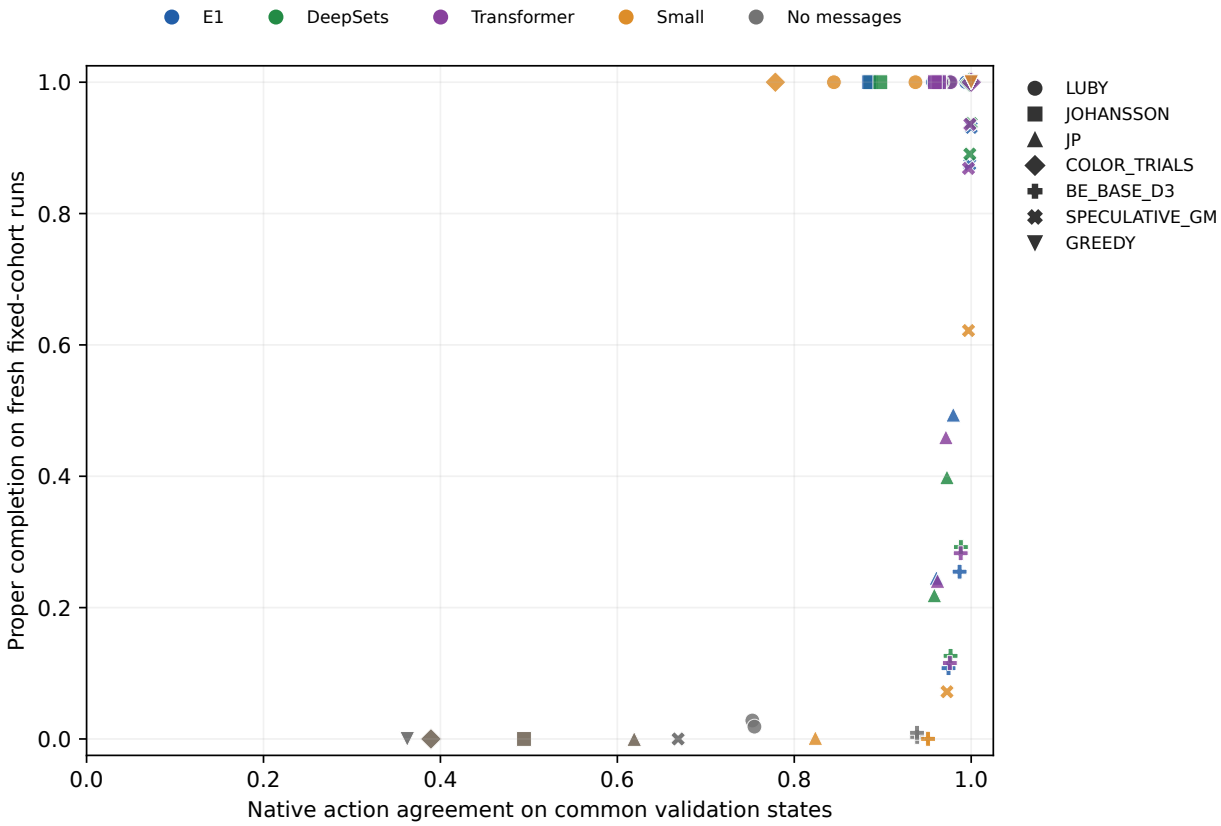


Figure 2: Common-state native action accuracy and autonomous proper-completion rate. Colors denote architectures and symbols teachers; all 70 checkpoints are plotted and identical points can overlap.

11 Complete-coloring laws and statistical reconciliation

11.1 Historical local-palette evidence

Table 5. Separate historical global-law cohorts. Counts are rejections after each declared Holm family. They are decisions under different cohorts, sample sizes and methods, not comparable numerical distances.

Architecture	Initial MMD:250 runs,265 comparisons	Robust V2:500 attempts,265 comparisons	Factorial V2:500 attempts,240 comparisons
E1	10	52	115
NORMALIZED			
DeepSets	46	107	170
Local palette	127	181	180
Transformer			

On identical new robust banks, the earlier single-scale MMD gives 21,71,147 rejections respectively; on identical factorial banks it gives 49,121,174. These matched-data comparisons isolate the procedure/budget difference from historical new-bank variation. They still do not turn rejection counts into a ranking of TV distances or make V2 uniformly more powerful. Archived auxiliary cases include losses as well as gains for V2.

The distributional differences coexist with strong functional generalization in the full-budget architecture cohort. Functional success restricts outputs to proper colorings but does not select a unique probability law on that set. The teacher itself need not sample proper colorings uniformly. Conversely, the clique symmetry argument gives genuine scoped equality despite different proposal mechanisms. Thus neither candidate mismatch nor success alone determines a global-law conclusion.

11.2 Reconciliation with the independent E1 behavior study

Pedro reports no rejections among 100 primary stratum/model/representation tests using five newly trained E1s, degree$\leq$ 3, common K4, cutoff 256 and 256 runs per graph. His experiment tests full labeled terminal vectors/status and a color-name quotient. It is not the same checkpoint cohort, degree/palette regime, cutoff or hypothesis family as Table 5. Ten labeled-law and eleven quotient-law pairs obtain simultaneous TV$\leq$ .05 certificates using coupling disagreement upper bounds. Those certificates are affirmative fixed-graph approximate-equivalence statements; zero rejections alone would not supply them. Their underlying raw banks and trained weights are not available here.

We independently verify the mixture statistic's equality to unbiased MMD and its inclusive permutation calculation. On identical archival 500-row banks from the first regular N128/degree 3 graph, neither mixture MMD nor V2 rejects any of the three s 41000 architecture comparisons. On the first regular N512/degree 100 graph, both reject all three. Each high-degree raw p-value is .00005; the exploratory six-test Holm value is .0003, whereas the original 720-test V2 adjusted value remains .036. The original statistics and p-values replay exactly. These are matched historical reanalyses, not new independent confirmation or a replay of Pedro's missing weights.

Different finite-sample power remains scientifically important. On a star with 200 iid binary leaves, change only the center from uniform on 199 high colors to uniform on a 100-color subset. Every vector is a proper local-list coloring, TV equals 99/199, and the mixture population MMD^2 is 9.569688e-6. Twenty fresh independent experiments per cell give Table 6.

Table 6. Independent matched-data controls, nominal alpha .05,999 permutations. Entries are detections out of 20 independent experiments.

Law	Runs per generator	Mixture MMD	Honest V2
Same-law star	256	0	1
Same-law star	500	1	1
Center-only alternative	256	0	20
Center-only alternative	500	3	20

For the alternative the pointwise exact 95% power intervals are[0,.1684] and[.0321,.3789] for the mixture and[.8316,1] for V2 at both sizes. Figure 3 preserves the null controls and uncertainty. This finite-law result does not establish strict Holm-adjusted power for 720 hypotheses or explain

every historical model comparison. In particular, it does not show that Pedro's degree-three neural laws exhibit this star alternative.

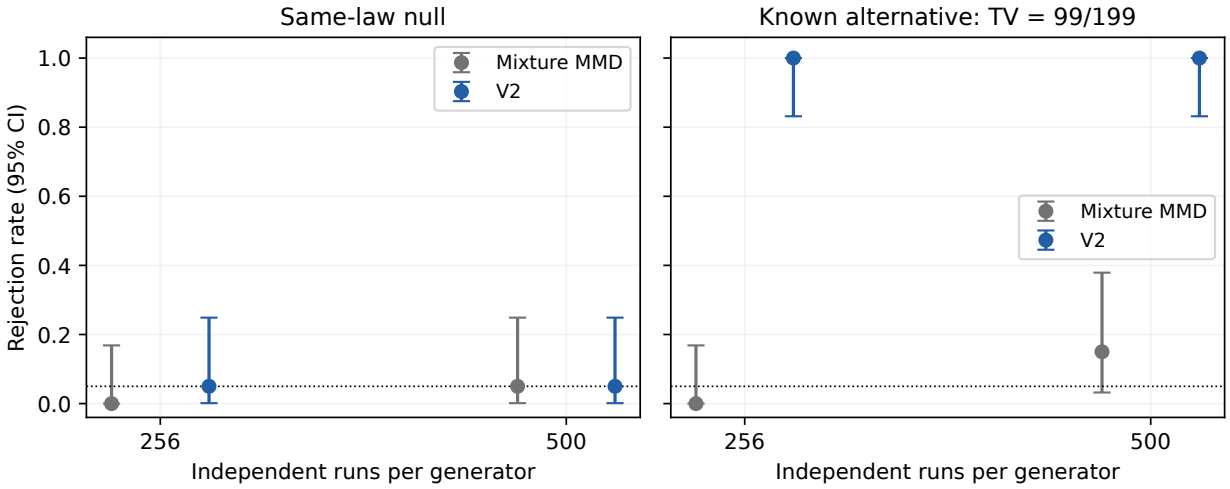


Figure 3: Independent known-law null and power checks. Both procedures receive identical fresh full-coloring rows in each trial; intervals are pointwise exact 95% binomial intervals over 20 experiments.

The discrepancy classification is consequently mixed: different output representations are different definitions; graph/palette/checkpoint/cutoff changes are protocol differences; the known-law power gap is statistical; and the effect of V2 on Pedro's exact original banks remains unresolved. No speculative explanation closes that final gap. The complete audit is `docs/integration/GLOBAL_DISTRIBUTION_RECONCILIATION.md`.

11.3 New teacher-specific global comparisons

Of 350 planned teacher/checkpoint/graph hypotheses, 227 are testable and 123 have fewer than four successful model/reference rows. After Holm over the complete planned family, 6 reject. Each reference bank belongs to the corresponding teacher. These counts concern conditional output laws on a small fixed cohort; functional failures and limited power remain separate.

All six new rejections concern two small checkpoints that nevertheless complete all 640 of their functional evaluations: LUBY seed 73001 and ColorTrials seed 73000. Each differs from its own teacher on three fixed graphs, with raw p .00005 and Holm-adjusted p .0175. The LUBY core has 551 parameters and the context-extended ColorTrials model 1,560. This supplies new finite-cohort evidence for functional/distributional separation under two teachers. The other small ColorTrials initialization fails all 640 functional attempts, so the successful checkpoint is not presented as universal small-model robustness. The large architectures have no new rejections here; those nonrejections do not certify their law equality.

The native comparisons use the audited V2 selection/permutation construction with an explicit global-K4 histogram alphabet. The historical feature allocator assumed max(degree)+1 bins, which is insufficient for valid four-color native path outputs. The first call failed before producing a p-value. A separate adapter repairs that allocation while preserving actual degrees, feature definitions and the original frozen module. An independent color 3 path-null test and a matching-alphabet exact replay verify the compatibility change. This is documented as an integration repair, not an

invalidation of the historical local-list study. The new native power is not inherited from its old validation by assertion.

Table 7. New Holm rejections / testable comparisons (ten planned per teacher and architecture).

Teacher	E1	DeepSets	Transformer	Small	No messages
Luby	0/10	0/10	0/10	3/10	0/2
Johansson	0/10	0/10	0/10	0/0	0/0
JP	0/8	0/7	0/7	0/0	0/0
ColorTrials	0/10	0/10	0/10	3/5	0/0
BE (base d 3)	0/6	0/7	0/5	0/0	0/1
Speculative GM	0/10	0/10	0/10	0/9	0/0
Greedy	0/10	0/10	0/10	0/10	0/0

Every law test conditions on proper native completion and compares with the corresponding teacher. The retained status/failure data are essential: a rarely successful network may have an estimable conditional law on some graphs while performing poorly overall. Untestable cells and nonrejections remain distinct. No test result establishes exact equality of all teacher and learner laws.

12 How size and degree affect distributional generalization

The factorial study varies n$\in$ {128,256,512,1024} and degree Delta$\in$ {3,10,50,100}, with three fresh regular graphs per cell and five fixed checkpoints per architecture. It uses local palettes,500 primary attempts and 125 separate explanation attempts per generator. Its target is this fixed graph/checkpoint cohort, not a population of infinitely many graphs or trainings.

Two preregistered events are $A = \{U(X) \leq \Delta\}$, with U the number of used colors, and $T = \{\text{color } \Delta \text{ occurs}\}$. EventA says that at least one palette color is absent, not necessarily the maximum. For E1 at degree 100, the equal-weight cohort differences in P(A), model minus teacher, are approximately 0,.157,.639,.603 as n increases through the four values. Their simultaneous intervals include[.570,.703] at n 512 and[.550,.629] at n 1024. At degree 50, the top-color difference instead attenuates from-.239 at n 128 to-.004 at n 1024. These are observable-probability effects, not a monotone full-TV scaling law.

Figure 4 displays the registered effects with their simultaneous bands for all three architectures at degree 100. Across the principal factorial, the E1 eventA interaction between n 256/n 512 and degrees 50/100 is .566 with simultaneous interval[.428,.704]. A shared teacher bank is counted once when averaging architecture effects. The earlier all-fifteen E1 replication at the new n 512/degree 100 graphs rejects 15/15 comparisons; combining the two prespecified event lower bounds gives a minimum simultaneous TV lower bound .078 across those fifteen graph/checkpoint pairs.

In regular graphs, degree also changes palette size. Their separate causal effects are not identified by this design. Comparing TV lower bounds on different graph spaces does not order the true TV values. Three graph instances and five historical trainings constrain population claims. The independent holdout witnesses and exploratory candidate interventions remain separate analyses; an observed proposal/final-law association is not alone a causal mediation result.

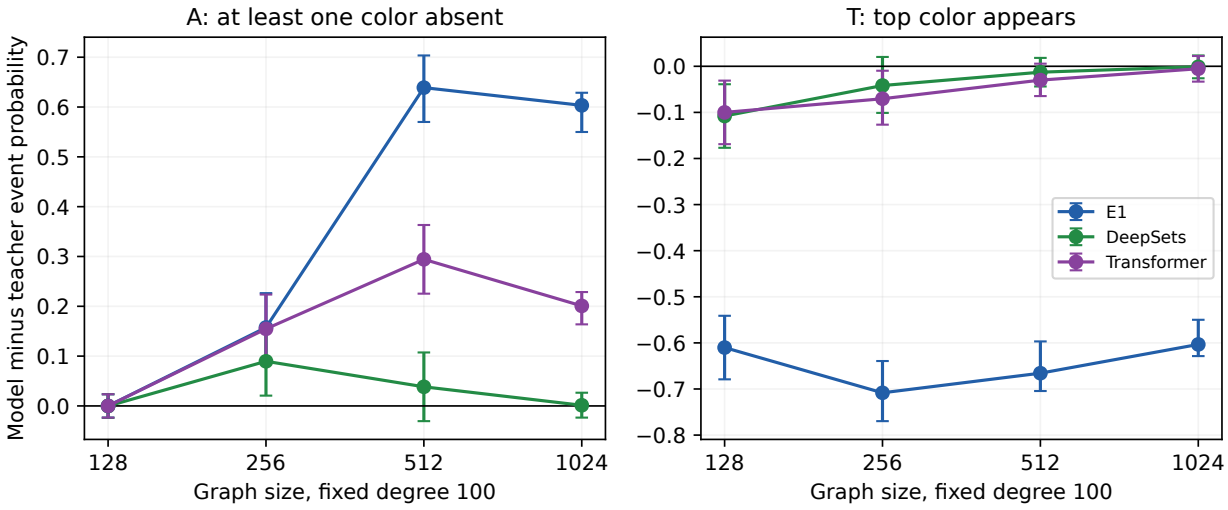


Figure 4: Fixed-cohort event-probability differences at degree 100, with simultaneous historical CP/Hoeffding bands. Curves concern two specific observables and do not order complete-law TV across graph sizes.

13 What the baselines establish about GNNs

Neighborhood information has a precise role: free-color decisions inspect fixed neighbors, and confirmation can require neighbors' availability-dependent offers. The scoped indistinguishability examples prove limits for an own-state or one-round current-state interface, rather than proving that a large GNN is necessary for graph coloring. Exact classical rules already solve the task with no trainable parameters when their required information and schedule are supplied.

MAX provides a degree-insensitive representation of Boolean existence and permits the finite-type COMMIT reduction. SUM can encode counts and MEAN proportions; a training procedure may exploit multiplicity-dependent shortcuts that fail outside the observed degrees. The historical negative controls demonstrate such risks for specific learned models. They do not constitute expressivity impossibility results for those aggregators.

DeepSets and attention investigate how a node represents its palette. Their success shows that the original prefix scorer is not the only sufficient mechanism in the tested setting. Both still communicate over graph edges. The small control tests reduced width within a normalized family. The no-communication control tests missing information while preserving core parameter count and supplied permissions. Any comparison of those baselines must retain these differences.

The small LUBY model has 551 trainable parameters and completes all 1,280 LUBY evaluations; the small greedy control also completes all 1,280 under its supplied schedule. Thus the larger core is not empirically necessary for those finite cells. In contrast, the no-communication control completes 38/8,960 total attempts across the seven teachers. This supports the value of received information under this protocol, without proving that the particular large GNN or a learned message MLP is necessary. Small-model failures for other teachers are retained and do not establish incapacity beyond this budget.

Teacher independence is similarly scoped. One architecture can expose the action alphabet for all seven teachers, but supplied preferences, priorities, phases and schedules change the observation

interface. Success across those interfaces is not learning arbitrary unseen algorithms from identical raw inputs. The finite experiment supplies evidence of adaptability, with cells that fail retained as limitations.

14 Discussion

The evidence supports the proposed distinction among notions of learning. The native JP example separates local action accuracy from autonomous validity. The historical palette study separates successful coloring from exact candidate imitation. Full-law tests and event bounds separate task validity from teacher-distribution fidelity. Clique symmetry and scoped coupling certificates show why the converse statement, that every imitating network must have a different global law, would also be false.

Teacher changes can affect a policy without changing its unrestricted output law. Exact synchronous JP with fixed distinct priorities gives the same first-fit solution as sequential greedy in descending priority order. If ID order is uniform, ascending and descending orders are both uniform, so uncapped JP and random-order greedy induce the same final-coloring law. Their common-ID executions and native costs can differ. Conditioning JP on a restrictive 128-round cap can also change that equality. This elementary observation prevents equating algorithm identity with solution-law identity.

The statistical reconciliation illustrates a complementary point. A characteristic kernel identifies laws in the population but can place a large TV difference in a weak finite-sample direction. Honest local witness selection can help in a specified alternative class while preserving conditional calibration. It can still miss rare or unrepresented high-order effects. Statistical instruments are part of the experiment, not neutral or universally powerful labels attached afterwards.

The strongest paper narrative therefore concerns the relationship between teacher specification, architectural information and the notion of fidelity. It does not require deciding that one research line was right and the other wrong. Differences caused by definitions should be stated; actual bugs should be repaired with evidence; remaining numerical gaps should remain open. This approach permits a defensible joint interpretation without erasing contradictory-looking historical records.

15 Limitations and unresolved claims

The principal historical limitation is missing collaborator runtime data: five retrained E1s,35 native E1s, source pools and original full coloring banks. Published tables and sources can be checked, but exact model-level reproduction has not been completed. The new 70-model cohort is not a substitute for those artifacts. Independent scientific review of this integrated draft remains pending.

The new balanced design has two initialization replicas,24 training graphs, one trajectory per graph and a much smaller exposure budget than the earlier full-budget studies. It does not identify an asymptotic architecture ranking or separate optimization from capacity in failed cells. Native teacher generalization above degree 3 or to local palettes is unsupported by this interface. The separate historical local-palette evidence must retain its own checkpoints and protocols.

Finite functional success does not prove universal neural safety, termination or complexity preservation. Canonical MAX certificates require precise algebraic/numerical assumptions and do not cover

every native context head. The simulator is synchronous, with supplied schedules and arithmetic for some teachers; distributed preprocessing and asynchronous deployment are not evaluated.

Output-law tests have finite power and resolution. Their nonrejections are not equality statements, while conditional solution laws do not determine failure or stopping-time laws. Multiple separate historical families do not give one 5% guarantee over their union. CP/Hoeffding/DKW bounds concern their stated finite cohort or observable families, not a graph population. No causal degree/palette separation or general monotone TV law is established.

16 Conclusions

Learning from an algorithmic teacher has several distinct targets. Our integrated coloring project shows how the teacher's information and action interface constrain architecture design, how strong local imitation can fail in autonomous execution, and how functional extrapolation can coexist with a changed complete-solution law. Exact teacher baselines and smaller controls prevent attributing classical task solvability to GNN complexity alone. Symmetry and coupling also identify settings where distributional agreement is possible despite different decision mechanisms.

These conclusions are conditional on the stated protocols and evidence. The repository preserves all cohorts, negative results, repaired failures, checkpoint identities and reproducibility boundaries. Completing historical collaborator artifact recovery and independent scientific review is necessary before presenting the entire joint project as independently reproduced or submission-ready.

17 Appendix A: teacher proofs and execution audit

For permanent trials, at most d(v)-d_U(v) distinct colors are fixed nearby, so both a four-color list at degree≤ 3 and a local d(v)+1 list satisfy $|A_v| \geq d_U(v) + 1$. Conditional independence gives

$$P(J_v = 1 \mid q) = p_v \sum_{c \in A_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)).$$

For LUBY, $p_u = 1/2$ and the center law is uniform on A_v. The chord inequality $\log(1 - x) \geq 2x \log(1/2)$ on[0,1/2], followed by Jensen for the exponential, gives $P(J_v = 1 \mid q) \geq (1/2)(1/2)^{d_U(v)/|A_v|} \geq 1/4$. For JOHANSSON every pending neighboring candidate has mass at most 1/2 on any color, because its list contains at least two colors. Thus its collision-free product is at least $2^{-d_U(v)} \geq 1/8$ here. A ColorTrials vertex wins some color whenever its preference for one available color equals 3 and all eligible neighbors' preferences are≤ 2, giving 27/256. Iteration of the conditional noncompletion bound and a vertex union bound prove almost-sure termination and logarithmic expected native rounds. These arguments refer to fresh independent mathematical random inputs; the documented finite-grid numerical simulation is a separate object.

For JP, distinct adjacent local maxima are impossible and a largest pending ID always exists. Availability and induction prove safety; at least one pending vertex is fixed each round, giving at most n uncapped rounds. The path 129 increasing-ID counterexample proves that 128 is not a universal fixed termination cap. To see equivalence with descending-order first fit, induct along the larger-neighbor dependency DAG. Every vertex sees the same already-colored higher-priority neighbors in both executions. Uniform ID order makes its reversal uniform too, proving the unrestricted

output-law identity with ascending random-order greedy. A24-permutation path 4 test checks the finite example, without replacing the argument.

For Linial initialization, encode auxiliary colors below B as base-p polynomials of degree$\leq$ r. Distinct adjacent auxiliaries give nonzero polynomial differences; each has at most r roots in the prime field. With p>3r, some point avoids all at most three neighboring equalities. Adjacent nodes choosing the same output pair would have equal values at the same point, a contradiction. The implemented fields 13 and 7 satisfy these inequalities for B65,536 and 169, producing 49 auxiliary colors.

For KW set $q = \lceil 4B/5 \rceil$. Retain auxiliaries below q. Give each old color $a \geq q$ the disjoint block $\{4(B-1-a), \ldots, 4(B-1-a)+3\}$. All blocks lie below q because $4(B-q) \leq q$. At most three neighbor colors forbid choices, so a free choice exists. Recoloring endpoints have disjoint blocks, and recolored nodes avoid retained neighbor colors. Properness is preserved while the bound decreases. From 49 it reaches 4 after 13 such steps, giving 15 with the two field steps.

During exact GM assignment, a later slot avoids all earlier assigned colors; only same-slot assignments can conflict. Detection marks the lower-ID endpoint of every such conflict, so the unmarked subgraph is proper. In sequential repair, four colors suffice and the last repaired endpoint of an edge avoids the other's final color. Thus the final coloring is proper. For maximum block width w and m marks, the supplied schedule uses 2w+m steps. Greedy similarly finishes after n first-fit steps. Neither proof turns a sequential scheduler into an anonymous LOCAL algorithm.

Independent scalar neighbor loops match 7,641 native transition choices over 103 unlabeled simple graphs of at most six vertices and degree$\leq$ 3, with two specified tapes per graph/teacher:1,442 complete executions with no observed failures. Eight explicit random-array comparisons additionally match canonical and native LUBY at every state. This is finite-domain evidence over the sampled IDs/tapes, not exhaustive universal verification. The missing simple-undirected input guard was reproduced by 21 failing cases and repaired without changing supported graph rules. Full traces,+pre/post regression files and proof assumptions are archived.

18 Appendix B: certificates and statistical guarantees

The canonical second-stage candidate-slot input has six legal neighbor types: fixed-other, fixed-same, pending/waiting/other-offer, pending/waiting/same-offer, pending/active/other-offer and pending/active/same-offer. Nonnegative shared messages and MAX reduce arbitrary multiplicities to 64 subsets. For an active pending center with one legal candidate slot, the own input is fixed. The fixed-same and pending/active/same-offer types exactly characterize rejection. Checking every strict sign therefore suffices in real arithmetic. Stored float 32 weights are interpreted as exact rational coefficients for interval analysis; floating-point error enclosures additionally require their stated IEEE rounding, overflow and precision assumptions. This does not certify arbitrary CUDA kernels or every context-extended teacher head.

For fixed n and gamma>0, the one-coordinate Hamming kernel matrix on any finite color alphabet is $(1-a)I + a11^T$, with $a = e^{-\gamma/n} \in (0,1)$. Its eigenvalues are strictly positive. The full-vector product matrix is a Kronecker product; restriction to proper colorings is a principal submatrix and stays positive definite. Hence distinct finite probability vectors have positive population MMD^2 . This proves identifiability, not good power for a given graph and m. A terminal-status delta gives separate positive-definite blocks; a quotient representation instead changes the observed space.

Conditional on independent witness-selection data, V2's evaluation statistic is fixed and the pooled scaling ignores source labels. Under equal conditional solution laws, independently sampled evaluation row labels are exchangeable given the pooled observations and group sizes. The inclusive rank of the observed statistic among uniform random whole-row permutations, including the original, yields the super-uniform plus-one p-value. Ties are conservative. The argument makes no independence assumption about vertices within a row. The declared-alphabet native histogram adapter preserves this proof because its features are deterministic and its selection split is unchanged.

For Holm, suppose m 0 hypotheses are true. If any true hypothesis is first rejected, its ordered p-value is at most alpha/m 0. A union bound on the m 0 super-uniform true-null p-values bounds this event by alpha, regardless of dependence between tests. This covers shared reference banks within each stated family, not an uncorrected union of multiple families.

The factorial event inference uses a saturated binomial probability per graph/generator bank. Architecture means give equal weights to five fixed checkpoints and three fixed graphs, retaining the shared teacher term once. Simultaneous Clopper–Pearson intervals receive total failure budget .025; weighted Hoeffding contrasts receive .025 across 1,800 prespecified contrasts. Their intersection has simultaneous coverage at least .95 by union bound under the stated independent-run fixed-bank model. These are event-risk contrasts, not confidence intervals for a population of graphs or a general TV scaling parameter.

19 Appendix C: evidence map and reproducibility boundaries

ID	Claim	Experiment	Evidence boundary
C01	Full-budget functional extrapolation	paper-study	Reported archive plus immutable artifact manifests; finite cohort only
C02	Native teacher comparison	Pedro native	Published aggregate evidence; original raw/weight reproduction incomplete
C03	Pedro E1 output-law findings	Pedro behavior	0/100 reported rejections; coupling certificates reported, raw verification pending
C04	Teacher correctness	Exact native rules	Conditional proofs plus 1,442 finite-domain runs; no universal learned-policy inference
C05	Matched full-law comparison	Six archived banks	Original V2 statistics replayed; exploratory historical reanalysis

ID	Claim	Experiment	Evidence boundary
C06	Known-law sensitivity gap	Fresh independent control	20 trials per law/size; pointwise CP; no universal power conclusion
C07	Balanced multiple-teacher learning	New 70-model cohort	Fresh banks, two initializations, short budget; fixed graphs, not population rankings
C08	New conditional global laws	New native K4 cohort	350 planned Holm comparisons; untestable cells distinct from equality
C09	Size/degree observable effects	Historical factorial	1,800 simultaneous fixed-cohort contrasts; degree and palette confounded
C10	Input guard repair	New regression	21 failures before, pass after; no supported historical result invalidated
C11	Declared alphabet repair	New integration compatibility	Failed before path p-value; exact matching-alphabet replay and null regression
C12	Architecture necessity limits	Specified observations	Scoped information arguments and finite sufficiency, no global minimality theorem

The repository's `docs/PAPER_SCIENTIFIC_AUDIT.md` supplies the substantive independent-review handoff, including claims that remain unsupported. Raw integration banks and all 70 final checkpoints accompany their SHA256 manifest. Historical recovery releases include the local-list banks and 85 selected paper-study checkpoints. The separately listed missing Pedro weights/banks prevent claiming full joint historical reproduction.

The earlier manuscript and historical result sections remain recoverable from preserved branches and the archived pre-integration manuscript. Code organization retains the `src/dcl/` package to preserve checkpoint and import compatibility. Plot/table generation is separate from training and sampling. Repeated analysis regenerates data-derived artifacts, not new favorable results. Exact commands and their tested scope are in the README.

Acknowledgments

The preserved Git history attributes the degree/palette and mechanism studies to Martín Ríos-Wilson and the native-teacher/retrained-E1 studies to Pedro Montealegre. This draft preserves that provenance without settling author order. Original algorithm authors and software dependencies retain their existing attributions. The repository does not invent an accepted publication or a license absent from the project.

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